

Final Project Deliverables

Introduction to Computational Mathematics

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Overview

#	Deliverable	Weight
1	Code	50%
2	Report	10%
3	Slides & Presentation	40%
4	Peer Evaluations	Multiplier

 **Submission:** Canvas

Sharing

- I will share your code and reports with the class (with code hidden).
- With your permission, I may feature your project on my website as a resource for future students.
- For outstanding projects, I will encourage you to continue development for submission to conferences, journals, or JHU Design Day.

Grading Philosophy

Deliverables are assessed on how rigorously they address the three core project elements:

Element	Focus
Learning	Expansion of your computational toolkit with new methods.
Analysis	Methodological rigor and testing of boundaries/stability.
Research	Implementation and evaluation of a creative modification.

Assessment Standards

- **Non-relative:** You are not competing against your peers; grades are absolute.
- **Execution-based:** I will not grade the “difficulty” of the topic, but rather the depth and clarity of the execution.

Code (50%)

- This is your **primary deliverable**.
- Your code must be **well-documented** and **reproducible**.
- Use a **literate programming** style: intersperse code blocks with Markdown to create a narrative of your discovery.

Requirements & AI Policy

- **Environment:** Must run without errors on a fresh environment.
- **AI Usage:** You are encouraged to use AI for docstrings, comments, and unit tests. However, you are strictly responsible for whatever you submit. “AI slop”—unverified, redundant, or incorrect code generated by a LLM—will result in significant point deductions.

You are responsible for everything you submit, whether generated by you or AI.

Code Checklist

- ☐ Code runs from start to finish on a clean kernel.
- ☐ Every function has a descriptive docstring.
- ☐ Variable names are intuitive (e.g., `step_size` vs `ss`).
- ☐ Visualizations include titles, axis labels, and legends.
- ☐ Narrative (markdown) cells explain the code.

Report (10%)

- The report should be a high-level summary of your findings, not a reproduction of the notebook.
- **Target Audience:** Your classmates. Assume they understand course concepts but not your specific project.
- **Constraint:** Do not include long derivations or code snippets unless they represent your unique research contribution.

Format & AI Policy

- **Length:** 2–5 pages (excluding figures; excluding references).
- **AI Usage:** While you may use AI to draft summaries from your notebook, you must audit the output for “hallucinations” or vague filler. The final text must be precise and technically accurate. Once again,

You are responsible for everything you submit, whether generated by you or AI.

Report Checklist

- ☐ Clear problem statement and motivation.
- ☐ Summary of algorithms and key parameters used.
- ☐ Explicitly identifies your “Delta” (the research contribution).
- ☐ Includes at least 2–3 foundational references.
- ☐ Submitted as a PDF.

Slides & Presentation (40%)

- Slides serve as an “advertisement” for your work.
- **Duration:** 5–15 minutes per group.
- **Focus:** Spend more time on the *problem* and *results* than the technical minutiae of the code.

Suggested Timing

Section	Time
Introduction & Motivation	4-5 min
Methods Overview	2-3 min
Your Contribution & Results	5-6 min
Conclusion & Q&A	2-3 min

Presentation Checklist

- ☐ Maximum of 15 slides.
- ☐ Visual-heavy design (minimal text/bullets).
- ☐ One clear “takeaway” plot or table.
- ☐ Rehearsed for timing (exceeding 15 minutes will significantly impact grade).

Common Sense Warning

Code Functionality: In past terms, I've had to fail projects because the submitted code was non-functional. I am not responsible for debugging your submission. **If the code does not run, the deliverable cannot be graded.**

Graduating Seniors: I am under no obligation to grant a passing grade for incomplete or non-working submissions. If anything, you are even more accountable to ensure your project is complete and functional, as it may impact your graduation status.

Peer Evaluations

To ensure equitable contribution, you will evaluate your partners at the conclusion of the project.

- **Criteria:** Participation, Technical Contribution, and Collaboration.
- **Impact:** Scores act as a **multiplier** on your final project grade.
- **Privacy:** All evaluations are strictly confidential and will not be shared with group members.

I will collect both quantitative ratings and qualitative feedback.

Deadlines & Submission

Deliverable	Due Date
Slides	May 04, 11:59 PM or
	May 07, 11:59 PM
Project Archive	May 08, 11:59 PM

Policies

- **Zero Tolerance:** Late submissions receive a grade of **zero**.
- **Safety Net:** Canvas allows unlimited resubmissions until the deadline.
- **Strategy:** Submit a “v1.0” early to guard against technical glitches; overwrite it later with your final version.

Naming & Packaging

0. Choose a group name! Your group number can be found on [Canvas > People > Final Project](#).
1. Name files: [GroupNum_Name_Code.ipynb](#), [GroupNum_Name_Report.pdf](#), etc.
2. Zip all files (except slides) into: [GroupNum_Name_FinalProject.zip](#).
3. Submit slides separately via the dedicated Canvas assignment.



Good luck! I look forward to seeing your projects.